

Circular by Construction: What Synthetic Process-Data Generators Can and Cannot Establish for Batch Release

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Abstract

Synthetic “digital twins” of batch manufacturing are almost always validated against the simulator that produced their training data. That practice supports some conclusions and not the one that matters. We build a two-layer open benchmark separating the ground-truth process from the learned twin—a mechanistic fed-batch simulator whose failure-mode prior is anchored to 17,898 FDA drug-recall records, and 6 generative models trained on its output—and run one protocol on both corpora. No *pharmaceutical* batch records are public, so the real corpus is SECOM (1,567 semiconductor production runs, 446 signals, 6.64% failing), a structural stand-in. Fidelity and privacy conclusions transfer: over 6 such metrics the per-seed rank correlation between corpora is $\rho \in [0.55, 0.97]$. Utility conclusions do not ($\rho = 0.09 \pm 0.31$ for transfer AUROC, 0.14 ± 0.35 for AUPRC; both intervals span zero). Absolute utility does not carry over either: at comparable prevalence the simulated task is far easier (train-on-real AUPRC 0.893 ± 0.037 against 0.157 ± 0.022). That gap belongs to *this* simulator, whose disposition label is near-deterministic in its own trajectory; a builder without real batch records cannot know its size or sign. Generators emit failing real batches at $0.02\times$ to $3.92\times$ the true rate while aggregate fidelity barely registers it, and trajectory monitoring detects 0.673 ± 0.066 of failures leaving a process signature against 0.136 ± 0.159 of those that do not. We offer a benchmark and a protocol, not a release-decision system, and say which parts stay with the quality unit.

This work was carried out independently, on personal time and equipment, and is not connected to the author’s employment. The views expressed are the author’s own and do not represent the views, positions or policies of any current, former or future employer or client. No proprietary, confidential or internal data of any organization was used. All data is public: the SECOM dataset from the UCI Machine Learning Repository, the openFDA drug enforcement (recall) bulk export, and a simulated corpus generated by the code released with this paper.

1 Introduction

A “digital twin” of a pharmaceutical manufacturing process is usually proposed with two applications in view: detecting abnormal process trajectories while a batch is running, and anticipating whether the batch will meet its specifications at release. Both are attractive, and both run into the same obstacle. Real GMP batch records are among the most tightly held data in the industry. There is no public corpus of them. Every openly reproducible study of batch monitoring or release prediction therefore works on simulated data.

That constraint is usually acknowledged and then treated as a nuisance. We argue it is the central methodological problem, and that its consequences are measurable. When a generative

model is trained on simulator output and then scored against further simulator output, the evaluation is closed: it establishes that the model learned the simulator, and says nothing about the process the simulator was meant to stand for. The literature on synthetic tabular data has developed careful fidelity and utility metrics [1], and has learned to test privacy claims rather than assume them [25]; but every one of those measures inherits whatever reference distribution it is given. Applied inside a closed loop they report high scores that mean nothing about reality.

The question we can actually answer with public data is narrower than “is the simulator realistic?” — which is unanswerable without real batch records — but it is not vacuous. It is: *do conclusions drawn from evaluating generators on simulated process data transfer to real process data?* Our answer turns out to depend on which conclusion. Judgements about how well a generator reproduces a distribution, and about whether it memorises its training records, transfer well. Judgements about whether the synthetic data is any *use* downstream do not transfer at all — and that is the judgment on which a release-support application would rest.

Contributions.

1. We state the circularity problem in digital-twin evaluation for batch release precisely, and give a two-layer benchmark design (Figure 1) that separates the ground-truth process from the learned twin so that the circularity is visible rather than implicit (§3, §4).
2. We release a mechanistic fed-batch simulator whose failure-mode prior is anchored to real data: 17,898 FDA drug-recall records (92.2% classified by an auditable rule set), of which the process-relevant subset is renormalized into the simulator’s fault model (§4.2). Two of its six failure modes deliberately leave no trajectory signature, so detection has a principled ceiling rather than an artificial one.
3. We run one identical fidelity/utility/privacy protocol on a real manufacturing corpus and on the simulated corpus, over 10 seeds, and report the agreement between them as the primary result: fidelity and privacy rankings transfer, utility rankings do not, and absolute utility levels are not comparable at all (§6.2).
4. We report the negative results rather than the best operating point: generators that reproduce the process well and the *failures* badly, a detection ceiling set by failure modes the instrumentation cannot see, and false-alarm inflation of roughly 7-fold from testing repeatedly along a batch — the last two measured on the simulated corpus only, for want of any real trajectories (§6).
5. We state the decision-support boundary explicitly: what a system of this kind could contribute to a release workflow, what stays with the quality unit, and what a qualification path would require before any of it could be used (§7).

We do not claim a release-readiness system. We claim a benchmark, a protocol, and evidence that the current evaluation practice does not support the claims made with it.

2 Batch release is not a prediction problem

It is worth being precise about the act this work is adjacent to, because the imprecision is common and it matters for what may be claimed.

In the United States, §211.22 of 21 CFR Part 211 vests in the quality control unit the responsibility and authority to approve or reject all components, drug product containers, closures, in-process materials, packaging material, labeling and drug products. §211.192 then requires that all drug product production and control records be reviewed *and approved by that unit* before a batch is released or distributed, and that any unexplained discrepancy or failure of a batch to meet its specifications be thoroughly investigated — notably, *whether or not the batch*

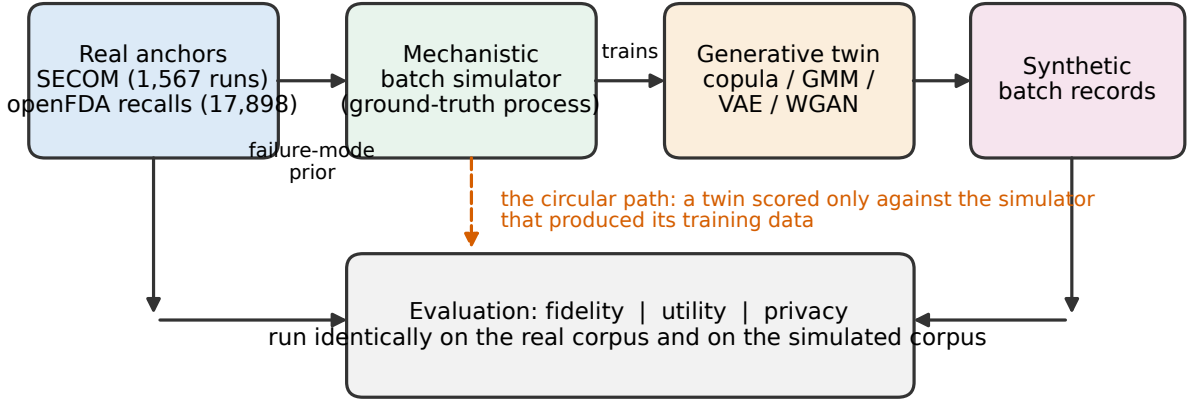


Figure 1: The two-layer benchmark. A mechanistic simulator plays the role of the ground-truth process; generative models play the role of the digital twin. The dashed red path is the circular evaluation this paper is about: a twin scored only against the simulator that produced its training data. The horizontal path is what we do instead — one evaluation protocol applied identically to the simulated corpus and to a real manufacturing corpus, so that the two answers can be compared.

has already been distributed [5]. In the European Union the corresponding act is certification by a Qualified Person under EudraLex Volume 4 Annex 16, which places personal responsibility on a named individual for confirming that each batch was manufactured and checked in compliance with the marketing authorisation and with GMP [7]. The two regimes differ in whom they name, but agree on the essential point: release is a documented human decision with an accountable owner, taken on a complete batch record, not an inference.

Nothing in this paper changes that, and no result below should be read as progress toward changing it. A model cannot hold the authority that §211.22 assigns or sign the certification Annex 16 requires. What a model can do is order work, focus attention, and flag records for review earlier than they would otherwise be seen. We return to exactly which of those is supported by our evidence in §7.

3 The circularity problem

Let $\mathcal{P}$ denote the real process distribution over batch records, and $\mathcal{S}$ a simulator intended to approximate it. A generative twin G is fitted to a sample $D \sim \mathcal{S}$ and produces synthetic records $\hat{D} \sim G$. The standard evaluation reports a divergence $d(\hat{D}, D')$ for a further $D' \sim \mathcal{S}$.

Two distinct quantities are being conflated. The reported $d(\hat{D}, D')$ bounds how well G learned $\mathcal{S}$. The quantity of interest is $d(\hat{D}, \mathcal{P})$, which by the triangle inequality is bounded below by $d(\mathcal{S}, \mathcal{P}) - d(\hat{D}, \mathcal{S})$. The term $d(\mathcal{S}, \mathcal{P})$ — simulator realism — is never estimated, because estimating it requires the real batch records whose absence motivated the simulator in the first place. It is not small by assumption, and no amount of improvement in $d(\hat{D}, \mathcal{S})$ constrains it. A twin can be arbitrarily good at reproducing a simulator that is arbitrarily wrong.

This is why we do not attempt to argue that our simulator is realistic. We instead test a weaker but checkable proposition. Write $\mathcal{R}$ for a real process distribution from which we do have a sample. If a metric’s ranking of generators computed under $\mathcal{S}$ agreed with its ranking under $\mathcal{R}$, then simulator-based evaluation would at least be a useful proxy for method selection, whatever its absolute bias. If the rankings disagree, the proxy fails on its own terms. §6.2 measures that agreement.

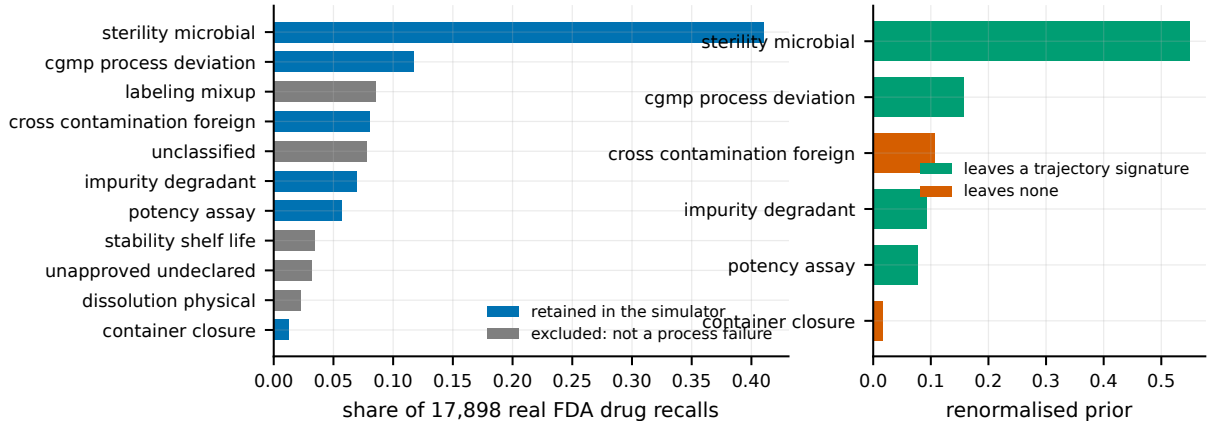


Figure 2: Left: empirical failure-mode distribution over 17,898 real FDA drug recall records. Modes that are not bioreactor process failures (labeling mix-ups, regulatory status, shelf-life, solid-dose attributes) and unclassified records are excluded. Right: the retained modes after renormalization, colored by whether the mode leaves a signature in the process trajectory. The two red modes set a hard ceiling on what any trajectory-based monitor can detect.

4 Benchmark

4.1 Real anchors

No public corpus of real pharmaceutical batch records exists, so we use two real corpora that are each partial substitutes, and we are explicit about which question each can and cannot answer.

SECOM: real process data with real disposition labels. The SECOM dataset [20] contains 1,567 production runs from a semiconductor fabrication line, each with 590 sensor measurements and a pass/fail disposition, collected over 89 days. 104 runs (6.64%) failed. The corpus has the pathologies of real plant data: 4.54% of cells are missing, one signal is missing in 91.2% of runs, and 116 signals are constant. After dropping constant and mostly-missing signals and median-imputing the rest, 446 signals remain. It is not pharmaceutical. It *is* a real, high-dimensional, weakly-labelled manufacturing process with a rare failure class and a genuine disposition decision attached, which is the structure our protocol needs. We use it to ask whether conclusions transfer, not to claim pharmaceutical relevance.

openFDA recalls: a real prior over failure modes. We use the openFDA drug enforcement corpus (export 2026-08-27) to anchor *which* failures occur and how severe they are. 17,898 drug recall records were classified into manufacturing failure modes by an ordered regular-expression rule set; 92.2% matched a rule and the remainder are reported as unclassified rather than forced. Microbial and sterility failures dominate at 41.1% of records, and 9.7% of all records are Class I. Figure 2 shows the empirical distribution and the renormalization into the simulator’s fault model.

This anchor has a bias that must be stated: recall records describe failures that *escaped* release and were detected in the market. Modes that batch release reliably catches are systematically under-represented. The corpus therefore constrains the failure-mode *mix* of the simulator; it cannot estimate a batch rejection rate, and we do not use it for one.

4.2 Mechanistic simulator

The ground-truth process layer is a fed-batch fermentation model in the Bajpai-Reuss tradition [2], of the kind used in the PenSim [3] and industrial-scale IndPenSim [10, 11] benchmarks.

Biomass, substrate and product follow Monod kinetics with dissolved-oxygen, pH and temperature dependence; production uses a lower half-saturation constant with substrate inhibition, so the process keeps producing under the substrate-limited regime it is deliberately operated in. Feed is exponential during the production phase; pH, temperature and dissolved oxygen are held by PI controllers with actuator noise. Seventeen process parameters vary batch to batch with stated coefficients of variation, representing seed-train and raw-material variability; Appendix A lists what they cover and how they were set.

Each batch runs 230 h and is sampled every 2 h across eight process variables, then passed through a sensor layer that adds measurement noise, a per-batch linear drift, and random dropouts. Batches are summarized into a 136-feature batch record: phase-wise mean, standard deviation, minimum and maximum of every variable over four process phases, plus eight release CQAs.

Failure modes and the observability ceiling. Six failure modes are injected with the renormalized recall prior. Four leave a trajectory signature (microbial contamination, chronic under-production, a thermal excursion driving degradant formation, a controlled-parameter deviation). Two do not: foreign particulate contamination and a container-closure defect arise after or outside the process the sensors observe. This is deliberate. A benchmark in which every fault is detectable rewards models for solving a problem that does not exist, and hides the fact that trajectory monitoring has a real upper bound set by what the instrumentation can see.

What this simulator does and does not claim. We make no realism claim for the simulator, and we avoid the word faithful, because as commonly used it admits no test. What can be checked about this model is its provenance and the behavior its equations entail: the model form is the published fed-batch structure cited above; production continues under substrate limitation with inhibition at high substrate; a competing organism depletes dissolved oxygen and substrate before product titer falls (Figure 3); and a thermal excursion raises degradant through the hydrolysis term. Those are refutable statements about this model, and a reader who finds the trajectories inconsistent with them has found an error. Nothing beyond them is asserted: no parameter was fitted to plant data, the absolute titer scale is arbitrary, and the model is best read as a parameterised data-generating process rather than a representation of any plant. Appendix A records how its parameters were set. We therefore do not assert real product specifications. Limits are derived from the simulator’s own nominal distribution over 500 fault-free batches, at the 0.27th and complementary percentile ($\pm 3\sigma$ equivalent). This is a direct consequence of having no real batch data to calibrate against, and it is exactly the kind of unfalsifiable modeling choice this paper is warning about; we make it visible rather than bury it.

Over 2,000 batches, 131 received an injected fault and 147 (7.35%) fell out of specification, of which 121 had an observable mode and 10 did not. The nominal batch failure rate was set to match SECOM’s observed failure rate so the two corpora are comparable in class balance; this is an assumption, not an estimate. Figure 3 shows nominal and contaminated trajectories.

5 Generators and evaluation protocol

5.1 Generators

We compare 6 generators spanning the families a synthetic-tabular-data study is expected to cover: an independence baseline that reproduces every marginal exactly and destroys all dependence; a shrinkage multivariate Gaussian [18]; a Gaussian copula with non-parametric margins; a diagonal Gaussian mixture with components chosen by BIC; an MLP variational

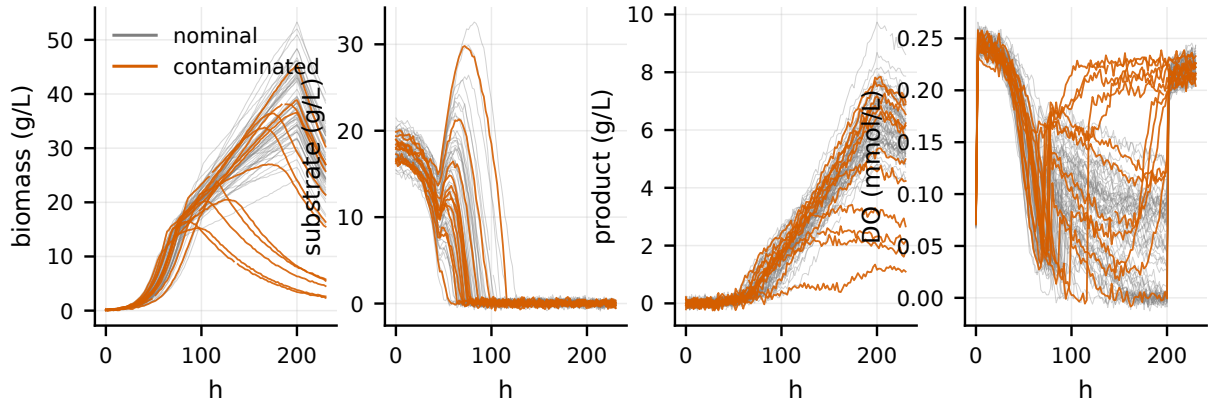


Figure 3: Simulated process trajectories: 60 nominal batches (grey) and 12 batches with microbial contamination (red). Contamination competes for substrate and oxygen, so its signature appears in dissolved oxygen and substrate before it appears in product titer — which is what makes early detection possible for this mode and impossible for the two modes that leave no trajectory signature at all.

autoencoder [17]; and a Wasserstein GAN with gradient penalty [12], standing in for the CTGAN family [28] without its categorical machinery, which this all-continuous setting does not require.

We additionally report *bootstrap resampling* of the training rows. This is not a method under evaluation. It is a reference point that is perfect on fidelity and utility and maximally bad on privacy, included so that the trade-off between the three axes is visible in the same table rather than asserted.

The disposition label is modelled jointly with the features, as a batch-record generator would have to, and thresholded at 0.5 when sampled.

Budgets. Reported so that a reader can judge whether a result is a property of a method or of its training budget. The VAE uses latent 32, two hidden layers of 256, 250 epochs, Adam at 10^{-3} , batch 128, with KL warm-up over the first 30% of epochs. The WGAN-GP uses latent 64, hidden 256, 1,200 generator steps, 3 critic steps per generator step, Adam at 2×10^{-4} , batch 128, gradient penalty 10. The Gaussian mixture selects $k \in \{2, 4, 8\}$ by BIC with diagonal covariance. The classifier two-sample test uses 3-fold cross-validated gradient boosting (150 iterations); the downstream classifier uses 200. Synthetic samples are drawn at $1 \times$ the training-set size for fidelity and privacy and $5 \times$ for utility. No per-corpus hyperparameter search was performed: the same budget is used on both corpora and for every seed.

5.2 Protocol

Each corpus is split into three disjoint halves-of-halves: a training set that fits the generator, a holdout that serves as the non-member set for privacy tests, and a test set used only for utility. No set serves two purposes. Per-generator results are reported as mean $\pm$ standard deviation over 10 seeds; rank correlations additionally carry a 5–95% percentile interval over the same seeds. No single-run number appears in this paper.

Fidelity. Per-column two-sample Kolmogorov–Smirnov statistics and Wasserstein distances for marginals; mean absolute difference of the Pearson and Spearman correlation matrices and top-10 PCA subspace overlap for joints; and a classifier two-sample test [19], cross-validated, where AUC 0.5 means indistinguishable.

Utility. Train-on-synthetic, test-on-real: a gradient-boosted classifier is fitted on synthetic records and evaluated on the held-out real test set, against a train-real/test-real baseline.

Because the failing class is rare, we draw five times the training-set size of synthetic records before fitting. A generator that still yields fewer than five failing records has failed to represent the minority class, and we report that outcome rather than working around it.

Privacy. Distance to closest record is reported as a *ratio* of median distance-to-training over median distance-to-holdout: a generator that merely produces typical records sits close to everything, so the absolute distance is uninformative and the ratio is not. A ratio near 1 means the generator is no closer to its training data than to unseen data of the same kind. We also report the nearest-neighbour distance ratio, the exact-copy fraction, and a membership inference attack in the sense of Shokri et al. [24], implemented here as a nearest-neighbour distance attack rather than their shadow-model classifier, and scored by AUC over training members versus holdout non-members.

6 Results

What we expected, and what we found. We state this before the numbers because it bears on how they should be read. The study was designed expecting that generator rankings would fail to transfer across the board, and the first draft of this paper asserted exactly that. They did not: fidelity and privacy rankings transfer well, and only utility fails. The framing in §6.2 was rewritten around that split after the results were in. No metric, generator or corpus was added or dropped in response to the outcome, and the analysis was run once over the ten seeds fixed in advance; but the *interpretation* is post hoc and we do not present it as a preregistered prediction. The simulator itself went through several calibration passes before any evaluation was run — kinetic parameters and the feed profile were adjusted until nominal titer and reject rate were plausible, and the specification limits were changed from asserted values to percentiles of the simulator’s own nominal distribution after the asserted values rejected every batch (§4.2). The neural budgets were also cut once for runtime, before the reported runs and applied uniformly thereafter: the WGAN from 3,000 to 1,200 generator steps and from five critic steps to three, the VAE from 400 to 250 epochs, and the classifier two-sample test from five folds to three. Both sequences are modeling history, not results, and they are disclosed here because a paper about undisclosed evaluation choices should not have any. §9 returns to what the reduced budgets may cost the neural generators.

6.1 Generator behavior on each corpus

Table 1 gives the full protocol on both corpora. Three patterns hold across both.

First, the classifier two-sample test saturates. On both corpora, most generators are separated from real data with AUC indistinguishable from 1.0: in several hundred dimensions with one to two thousand rows, a gradient-boosted classifier finds a discriminating direction for nearly any generator. C2ST is the metric a generative-modeling reviewer will look for, and on data of this shape it is close to useless for ranking. We report it because its saturation is itself a result about evaluating synthetic manufacturing data, not because it separates methods.

Second, marginal fidelity and joint fidelity come apart sharply. The independence baseline reproduces every marginal essentially exactly — by construction — and is among the best generators on mean KS while being the worst possible model of the joint distribution. Any evaluation that leads on marginal agreement will rank a model that has learned nothing about the process at the top.

Third, the bootstrap reference behaves as designed and makes the privacy axis legible: DCR ratio 0.000 ± 0.000 and membership-inference AUC 0.813 ± 0.016 on the real corpus, with an exact-copy fraction of 0.593 ± 0.029 . It is the best available generator on utility and offers no privacy at all. Every other row in the table should be read against it. Its C2ST AUC

Table 1: Fidelity, utility and privacy on both corpora, mean $\pm$ sd over 10 seeds. C2ST AUC 0.5 is ideal; the bootstrap row’s sub-chance C2ST is the duplicate-record artefact explained in §6 and is not comparable with the others. KS and correlation MAE lower is better; PCA overlap and TSTR AUPRC higher is better; DCR ratio 1 means no memorisation and MIA AUC 0.5 means the attack learns nothing. [†]Bootstrap is a reference point, not a method: it copies training rows.

corpus	generator	C2ST AUC	KS	corr MAE	PCA ovl	TSTR AUROC	TSTR AUPRC	DCR ratio	MIA AUC
SECOM (real)	<i>bootstrap</i> [†]	0.308 $\pm$ 0.024	0.029 $\pm$ 0.001	0.028 $\pm$ 0.003	0.717 $\pm$ 0.038	0.661 $\pm$ 0.044	0.145 $\pm$ 0.030	0.000 $\pm$ 0.000	0.813 $\pm$ 0.016
	marginal	1.000 $\pm$ 0.000	0.029 $\pm$ 0.000	0.059 $\pm$ 0.001	0.028 $\pm$ 0.005	0.468 $\pm$ 0.038	0.069 $\pm$ 0.016	0.992 $\pm$ 0.007	0.514 $\pm$ 0.017
	gaussian	1.000 $\pm$ 0.000	0.190 $\pm$ 0.005	0.037 $\pm$ 0.001	0.781 $\pm$ 0.024	0.533 $\pm$ 0.066	0.089 $\pm$ 0.025	0.986 $\pm$ 0.004	0.531 $\pm$ 0.018
	copula	0.957 $\pm$ 0.018	0.030 $\pm$ 0.000	0.044 $\pm$ 0.001	0.465 $\pm$ 0.038	0.554 $\pm$ 0.044	0.106 $\pm$ 0.024	0.980 $\pm$ 0.005	0.529 $\pm$ 0.016
	gmm	1.000 $\pm$ 0.000	0.132 $\pm$ 0.004	0.049 $\pm$ 0.004	0.391 $\pm$ 0.089	0.534 $\pm$ 0.052	0.092 $\pm$ 0.026	0.986 $\pm$ 0.005	0.520 $\pm$ 0.021
	vae	1.000 $\pm$ 0.000	0.192 $\pm$ 0.005	0.054 $\pm$ 0.001	0.470 $\pm$ 0.048	0.563 $\pm$ 0.057	0.116 $\pm$ 0.048	0.961 $\pm$ 0.009	0.592 $\pm$ 0.020
	wgan	1.000 $\pm$ 0.001	0.233 $\pm$ 0.008	0.068 $\pm$ 0.001	0.608 $\pm$ 0.036	0.566 $\pm$ 0.074	0.099 $\pm$ 0.019	0.975 $\pm$ 0.006	0.539 $\pm$ 0.018
simulated	<i>bootstrap</i> [†]	0.305 $\pm$ 0.012	0.028 $\pm$ 0.002	0.024 $\pm$ 0.001	0.842 $\pm$ 0.036	0.993 $\pm$ 0.005	0.950 $\pm$ 0.029	0.000 $\pm$ 0.000	0.810 $\pm$ 0.013
	marginal	1.000 $\pm$ 0.000	0.026 $\pm$ 0.001	0.116 $\pm$ 0.001	0.084 $\pm$ 0.016	0.412 $\pm$ 0.061	0.068 $\pm$ 0.016	0.985 $\pm$ 0.004	0.503 $\pm$ 0.022
	gaussian	1.000 $\pm$ 0.000	0.094 $\pm$ 0.003	0.025 $\pm$ 0.002	0.890 $\pm$ 0.028	0.780 $\pm$ 0.060	0.508 $\pm$ 0.095	0.972 $\pm$ 0.005	0.519 $\pm$ 0.024
	copula	0.841 $\pm$ 0.027	0.028 $\pm$ 0.003	0.035 $\pm$ 0.001	0.705 $\pm$ 0.021	0.704 $\pm$ 0.074	0.248 $\pm$ 0.087	0.969 $\pm$ 0.005	0.518 $\pm$ 0.025
	gmm	1.000 $\pm$ 0.000	0.051 $\pm$ 0.001	0.065 $\pm$ 0.001	0.457 $\pm$ 0.025	0.840 $\pm$ 0.042	0.613 $\pm$ 0.069	0.978 $\pm$ 0.006	0.515 $\pm$ 0.025
	vae	1.000 $\pm$ 0.000	0.104 $\pm$ 0.002	0.046 $\pm$ 0.004	0.699 $\pm$ 0.021	0.766 $\pm$ 0.115	0.512 $\pm$ 0.177	0.942 $\pm$ 0.008	0.579 $\pm$ 0.023
	wgan	1.000 $\pm$ 0.000	0.162 $\pm$ 0.008	0.076 $\pm$ 0.001	0.603 $\pm$ 0.021	0.724 $\pm$ 0.078	0.330 $\pm$ 0.108	0.968 $\pm$ 0.005	0.512 $\pm$ 0.026

of 0.308 ± 0.024 falls *below* chance, which is an artefact worth naming: because bootstrap duplicates training rows, the same record appears in both classes, and a classifier that memorises it in training is systematically wrong on it in validation.

On the metrics that do separate methods, the two corpora agree on the winner more often than not. The Gaussian copula is the only generator the C2ST cannot saturate on either corpus (0.957 ± 0.018 real, 0.841 ± 0.027 simulated), and the shrinkage Gaussian gives the best joint structure on both (correlation MAE 0.037 ± 0.001 real, 0.025 ± 0.002 simulated). It is on the third axis that the agreement breaks down.

6.2 Do conclusions transfer from simulated to real data?

This is the paper’s primary question. For each metric we rank the 6 generators by their mean on the simulated corpus and by their mean on the real corpus, and compare the two rankings. The answer splits cleanly by axis.

We report the correlation *per seed*: for each seed we rank the six generators on the simulated corpus and on the real corpus at that seed and correlate the two rankings, giving 10 estimates per metric rather than one point estimate over six method means. This matters, because with $n = 6$ a single Spearman coefficient carries no usable uncertainty and the paper’s claim is precisely a difference between coefficients.

Fidelity and privacy rankings transfer. Across the 6 fidelity and privacy metrics the per-seed correlation lies in $[0.55, 0.97]$, with every 5–95% interval clear of zero: marginal agreement $\rho = 0.97 \pm 0.04$, joint structure $\rho = 0.88 \pm 0.04$, memorisation ratio $\rho = 0.88 \pm 0.08$, and even the saturated C2ST reaching $\rho = 0.68 \pm 0.11$. If the only question were “which generator best reproduces this distribution, and does it copy its training records”, evaluating on a simulator would be a defensible proxy. That is a real and, to us, unexpected positive result.

One qualification belongs here rather than in the limitations. Agreement is only meaningful where the methods actually differ. Table 2 reports, alongside each correlation, the between-method spread divided by the seed-level noise. For the fidelity metrics that ratio is large (22.6 to 18.2); for the privacy metrics it is much smaller (as low as 1.1), meaning the privacy rankings are computed over methods that barely differ. We would not lean on the privacy transfer result on its own.

Utility rankings do not transfer. On the axis that a release-support application would depend on, the correlation collapses to $\rho = 0.09 \pm 0.31$ for transfer AUROC (5–95% interval

Table 2: Agreement between the two corpora. For each metric and each of the 10 seeds, the 6 generators are ranked on each corpus and the two rankings correlated, giving a mean $\pm$ sd and a 5–95% interval. “sep.” is the between-method spread divided by the seed-level noise, the smaller of the two corpora: a correlation computed where this is near 1 is a ranking of noise. The last column names the method each corpus selects; a check mark marks agreement on the winner.

axis	metric	ρ (per seed)	5–95%	sep.	selected method (sim / real)
fidelity	C2ST AUC	0.68 ± 0.11	[0.50, 0.82]	5.5	copula / copula ✓
fidelity	KS	0.97 ± 0.04	[0.91, 1.00]	17.8	marginal / marginal ✓
fidelity	corr MAE	0.88 ± 0.04	[0.82, 0.92]	6.9	gaussian / gaussian ✓
fidelity	PCA ovl	0.71 ± 0.14	[0.49, 0.83]	6.3	gaussian / gaussian ✓
utility	TSTR AUROC	0.09 ± 0.31	[-0.13, 0.47]	0.7	gmm / wgan
utility	TSTR AUPRC	0.14 ± 0.35	[-0.09, 0.57]	0.6	gmm / vae
privacy	DCR ratio	0.88 ± 0.08	[0.77, 0.97]	1.8	marginal / marginal ✓
privacy	MIA AUC	0.55 ± 0.22	[0.24, 0.80]	1.1	marginal / marginal ✓

[-0.13, 0.47]) and $\rho = 0.14 \pm 0.35$ for AUPRC ([-0.09, 0.57]). Both intervals contain zero and both contain negative values. The simulated corpus selects gmm (AUPRC 0.613 ± 0.069); the real corpus selects vae (0.116 ± 0.048 , on only 4 of 10 seeds — see §6.3). Choosing a generator by its downstream usefulness on simulated data gives approximately no information about its downstream usefulness on real data.

Comparing the two groups of per-seed correlations directly — 60 fidelity and privacy estimates against 8 utility estimates — gives medians of 0.83 and -0.03 (Mann–Whitney $p < 0.001$). We report this as a descriptive contrast rather than a confirmatory test, since per-seed correlations within a metric share the same six methods and are not independent in the strict sense.

Absolute utility is not comparable either, and the reason is worse. The two benchmarks are not equally hard. Training on *real* data and testing on real data — the ceiling each corpus offers, with no synthetic data involved — gives AUPRC 0.893 ± 0.037 on the simulated corpus against 0.157 ± 0.022 on the real corpus, at closely comparable prevalence (0.074 against 0.066). The simulated prediction task is nearly saturated while the real one is barely above its prevalence baseline. A synthetic-data method evaluated on the simulator is therefore being measured in a regime that does not exist in the plant, and any absolute utility number carried over from it is meaningless. Note that this is not a failure of the generators: relative to each corpus’s own ceiling, the best method retains 0.69 of it on the simulated corpus and 0.74 on the real one. What differs is the ceiling.

Table 2 gives the correlation for every metric together with which method each corpus selects, and Figure 4 plots the two corpora against each other metric by metric. Points on the diagonal would mean a generator scores the same on simulated and real data; a monotone arrangement would mean that even where levels differ, the ordering survives. The first holds almost nowhere — levels differ throughout — and the second holds on the fidelity and privacy panels and fails on the utility panel.

The finding is that contrast between axes, and it is large and consistent: every fidelity and privacy metric lands at $\rho \geq 0.55$ with an interval clear of zero, while both utility metrics land at $\rho \leq 0.14$ with intervals spanning it. Simulator-based evaluation is a reasonable proxy for distributional questions about a generator, and not a proxy at all for the downstream question of whether its output is useful.

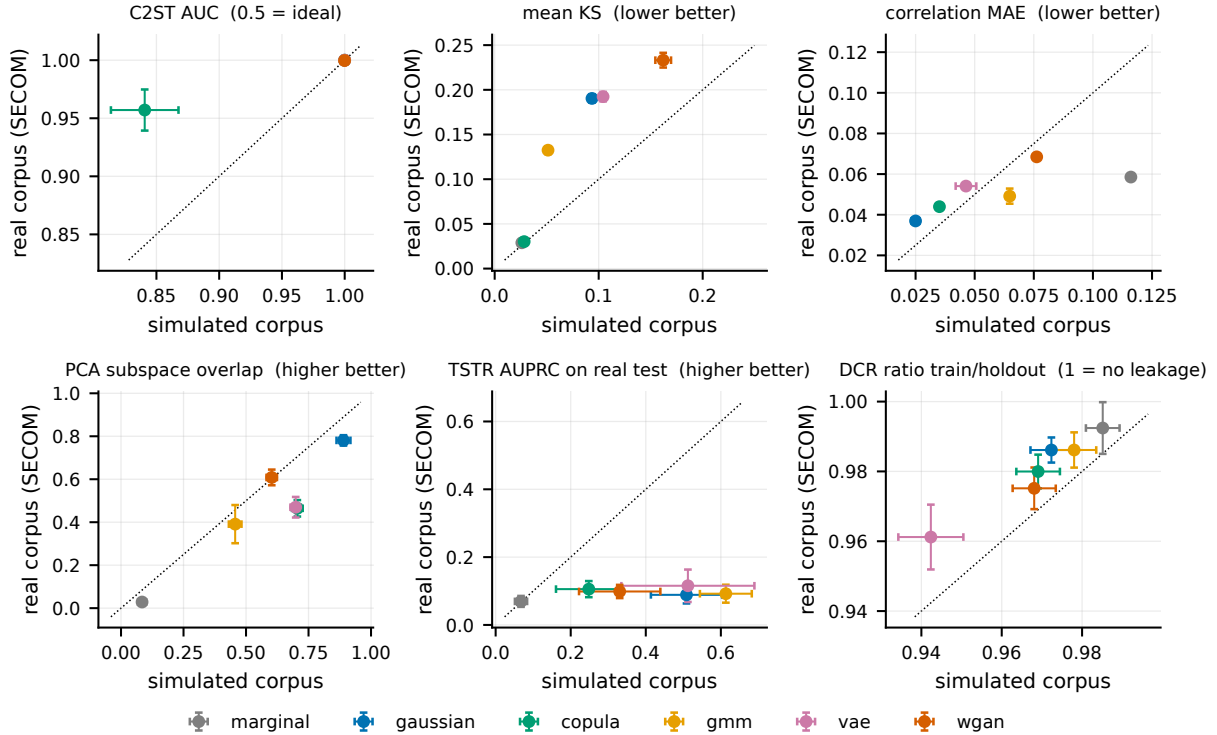


Figure 4: Each panel plots one metric, simulated corpus against real corpus, one point per generator, error bars are ± 1 sd over 10 seeds. The dotted line is $y = x$. Agreement in *level* would put points on the line; agreement in *ranking* — the weaker and more useful property — would make the points monotone. Levels differ on every panel. Ranking survives on the four fidelity and privacy panels and collapses on the two utility panels, which is the paper’s central result in one picture.

6.3 Generators reproduce the process well and the failures badly

The result with the most direct bearing on the application is the simplest. A synthetic batch-record corpus exists so that people can study batch *failure*. Table 1 scores generators on all columns at once, and on that basis several look respectable. Their reproduction of the failing class does not match.

On the real corpus, against a true failure rate of 0.066, the variational autoencoder generates failing records at 0.02 times the true rate and the WGAN at 0.09 times, while the shrinkage Gaussian generates them at 3.92 times — an error of one to two orders of magnitude in one direction and a fourfold excess in the other. The VAE’s deficit is severe enough that in 60% of seeds it produced too few failing records to fit a classifier at all, which is why its utility figure rests on 4 of 10 seeds and should not be compared with the others at face value. The copula and the independence baseline, by contrast, track the true rate closely (0.95 and 1.00 times) — the latter trivially, since it resamples the label column on its own.

None of this is visible in the aggregate metrics. A rate error of this size moves a single column out of several hundred, so mean KS, correlation MAE and C2ST barely register it. We therefore recommend that the minority-class rate be reported as a first-class metric in any synthetic manufacturing-data study, and we note that it is the one metric here that a practitioner could check in a minute without any of the machinery in this paper.

6.4 Privacy and the three-way trade-off

Figure 5 places fidelity against privacy on both corpora, with the bootstrap reference marked. The generators cluster near a DCR ratio of 1, consistent with sampling from a learned distribution rather than copying training records, while bootstrap sits at the degenerate corner.

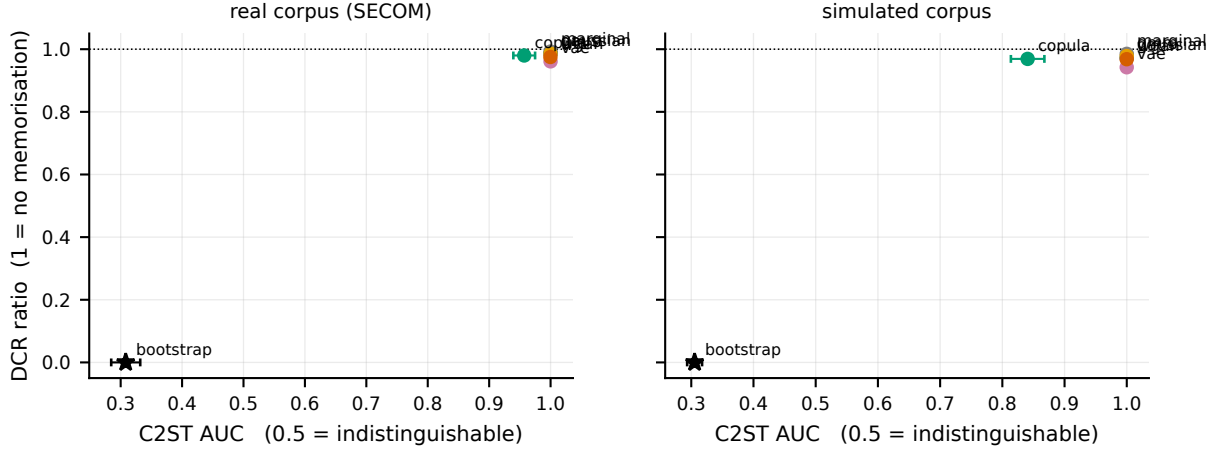


Figure 5: Fidelity against privacy, real corpus (left) and simulated corpus (right). Bootstrap ($\star$) copies training rows: it is the fidelity optimum and the privacy failure, and it fixes the scale for reading every other point. Error bars are ± 1 sd over 10 seeds.

Membership inference AUC is reported in Table 1.

We are careful about what this supports. A DCR ratio near 1 and a near-chance distance-based attack are evidence against *crude* memorisation, and nothing more. Stronger attacks with auxiliary knowledge defeat exactly this kind of check [25], and the corpora here contain no personal data, so these numbers are a methodological demonstration of the third axis rather than a privacy claim about any deployment. In a real setting the data at risk would be commercially confidential process knowledge, which has a different threat model than the individual-record disclosure these metrics were built for.

6.5 Monitoring and release readiness on the simulated corpus

The monitoring and readiness experiments can only be run on simulated data, since no real batch trajectories are available to this study. Every number in this subsection describes the simulator.

We want to be blunt about what that means, because it is the limit of our own escape from the problem this paper is about. We broke the loop for the tabular batch-record experiments by bringing in a real corpus. For the trajectory experiments below we did not, and could not: they are exactly as self-referential as the practice we criticise in §3, and §6.2 gives direct evidence that performance measured this way does not carry over to real process data. The numbers below should be read as a characterisation of the benchmark, not as an estimate of what a plant would see.

We use batch-wise unfolded multiway PCA in the Nomikos–MacGregor tradition [21, 22]: a PCA model is fitted on nominal training batches, and Hotelling’s T^2 and squared prediction error are evaluated on the partial trajectory at 8-hour intervals against limits calibrated on the nominal reference.

Table 3 and Figure 6 give the operating curve. Three things are worth stating plainly.

The nominal per-test level does not control the batch-level false-alarm rate. At a nominal $\alpha = 0.010$ the realised false-alarm rate on nominal batches is 0.070 ± 0.028 — roughly sevenfold — because two statistics are tested at each of 28 checkpoints along a batch. Any deployment of this method needs a multiplicity correction, and reporting the nominal level as if it were the achieved rate would overstate performance substantially.

At that operating point, failures with a trajectory signature are detected at 0.673 ± 0.066 , with a median time to first alarm of 136 ± 11 h into a 230-hour batch. Failures without a trajectory signature are detected at 0.136 ± 0.159 , which is not distinguishable from the false-

Table 3: Multiway-PCA monitoring on the simulated corpus, mean $\pm$ sd over 10 seeds. Detection is separated by whether the failure mode leaves a trajectory signature. The gap between the nominal α and the achieved false-alarm rate is the multiplicity effect of testing repeatedly along a batch.

nominal α	achieved false-alarm rate	detection: signature	detection: none	median TTD (h)
0.001	0.009 ± 0.003	0.348 ± 0.115	0.000 ± 0.000	124 ± 47
0.005	0.026 ± 0.007	0.552 ± 0.077	0.045 ± 0.096	150 ± 18
0.010	0.070 ± 0.028	0.673 ± 0.066	0.136 ± 0.159	136 ± 11
0.020	0.177 ± 0.026	0.727 ± 0.057	0.148 ± 0.163	124 ± 11
0.050	0.365 ± 0.018	0.797 ± 0.054	0.512 ± 0.288	90 ± 16
0.100	0.559 ± 0.015	0.885 ± 0.052	0.681 ± 0.178	55 ± 14

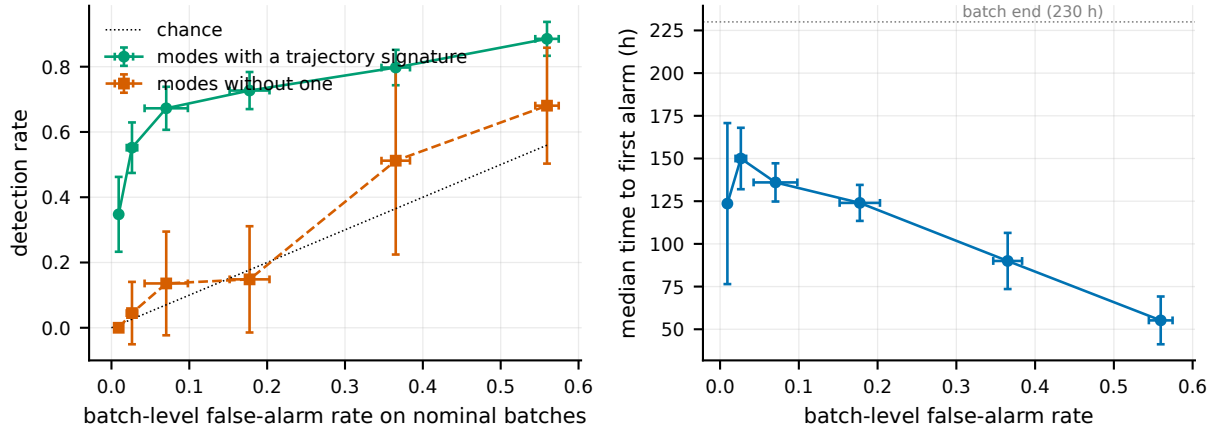


Figure 6: Left: detection rate against achieved batch-level false-alarm rate, separately for failure modes with and without a trajectory signature. The signature-free modes track the chance diagonal — the monitor is not detecting them. Right: median time to first alarm; earlier detection is bought entirely with false alarms.

alarm rate: the monitor is not detecting them, it is alarming at its base rate. The two arms rest on very different denominators — a test split holds 39 failures with a signature and only 4 without one, which is why the second figure carries a spread larger than itself. We report it because the argument is about observability rather than precision, but it should not be read as a well-estimated rate. No modeling improvement can change this, because the information is not in the data.

For end-of-production release readiness, a classifier on the trajectory up to 200 h reaches AUROC 0.799 ± 0.029 and AUPRC 0.612 ± 0.044 against a prevalence baseline of 0.074, a lift of $8.3 \pm 0.6\times$. That is a real signal on this simulator. It is also, and only, a statement about this simulator.

7 What this supports, and what stays with the quality unit

Given the results above, we state the proposed role narrowly.

What is supported. The *function* a monitor of this kind could serve is triage: ordering which batch records a reviewer opens first and which in-process investigations start earlier. We deliberately do not quote our readiness lift as the expected size of that benefit. It was measured on the simulator, and §6.2 is direct evidence that downstream performance measured on a simulator does not transfer. What our results support is that the triage function is worth investigating on real data, and nothing about how well it would work.

What is not supported, by our own evidence. Any use in which the model’s output substitutes for, shortens, or provides evidence within the release decision. Four results forbid it.

Downstream utility conclusions drawn on simulated data do not transfer to real data ($\rho = 0.14 \pm 0.35$, with an interval spanning zero; §6.2), so a system whose usefulness was established on a simulator has not had its usefulness established. The prediction task on our simulated corpus is also far easier than the real one (0.893 ± 0.037 against 0.157 ± 0.022 at comparable prevalence). We do not claim that magnitude is general — it follows from a label our simulator generates near-deterministically — and that is the point: a builder with no real batch records cannot establish whether their simulated task is easier than the plant’s or harder, so a performance figure carried over from simulation cannot be assumed to describe what the plant would see, in either direction.

A third of the failure modes in our fault model leave no trajectory signature and are detected at the monitor’s base rate, so a “no alarm” output carries no information about whole classes of failure. A negative result that cannot distinguish absence of fault from absence of observability must never be allowed to reduce scrutiny — this is the single most important sentence in this section for anyone tempted to deploy such a monitor.

And the specification limits themselves are derived from the simulator’s own nominal distribution (§4.2), so “predicting release readiness” here means predicting a label the simulator invented.

What stays with the human. All of it. The disposition decision, the review of the complete batch record required by §211.192, the investigation of any specification failure or unexplained discrepancy, and the certification signature under Annex 16 remain with the quality control unit and the Qualified Person respectively [5, 7]. A model output is at most an additional input to a human review that would happen anyway, and its absence of an alarm must not shorten that review.

The EU is drafting a rule that would exclude this model class outright. Annex 22 of EudraLex Volume 4, the GMP annex on artificial intelligence, is the instrument most directly relevant to everything above. It is a draft issued for consultation and is *not* in force: the adopted annexes of Volume 4 run to Annex 21, and Annex 22 is not among them [8]. What it proposes matters here anyway, because it is narrower than the framing of this paper. Its scope covers models “used in critical applications with direct impact on patient safety, product quality or data integrity”, and it applies only “to static models, i.e. models that do not adapt their performance during use by incorporating new data”. Dynamic models “should not be used in critical GMP applications”, and the draft states plainly that it “does not apply to Generative AI and Large Language Models (LLM), and such models should not be used in critical GMP applications”.

Every generator studied in this paper is a generative model. If Annex 22 is adopted in anything like its draft form, the model class evaluated here is excluded from critical GMP applications by rule, independently of how well it performs. That does not make this work moot — a benchmark, an evaluation protocol and a synthetic corpus for research are not critical GMP applications, and the draft does not reach them. It does mean the honest reading of our results is narrower still than the transfer result of §6.2 already makes it: not merely that the evidence does not yet support a release-decision role, but that the regulator with jurisdiction is drafting a rule which would forbid one for models of this kind. We report this as the current state of a draft, not as settled law.

The qualification path this has not started. “Future work” understates the distance, so rather than name three frameworks we work the opening steps of one of them through concretely.

FDA’s draft credibility-assessment approach for AI supporting regulatory decisions proceeds through a sequence of steps that begins by defining the question of interest, then the context of use, then assessing model risk, before any credibility evidence is planned or gathered [26]. Those first steps are the only ones a paper without real batch data could discharge, and attempting them is instructive.

The question of interest is the easy part: *does this batch conform to its specification?* The context of use is where the exercise breaks. A context of use has to state which decision the model informs, what else informs that decision, and what the model does not cover. Written honestly for the system studied here it reads: a multiway-PCA monitor, trained on synthetic batch records from a simulator of unestablished realism, ordering the queue in which a reviewer opens batch records for a process, product and site the model has never seen, with no coverage of two of the six failure modes in its own fault model and none of any mode absent from that model. That is narrow to the point of being nearly vacuous, and writing it down is what makes it obvious — which is the argument for drafting a context of use at the start of such a project rather than at its qualification.

Model risk, in the draft’s terms, is a function of how much the model influences the decision and how severe the consequence of a wrong decision is. Here the consequence is the most severe the framework contemplates: a non-conforming batch released to a patient. Influence is therefore the only available lever, and the triage framing above exists to hold it near zero — the model changes the *order* of a review that happens in full regardless, and no model output shortens, substitutes for, or contributes evidence to the disposition. Severe consequence with near-zero influence is the only cell of that matrix this system can occupy today. That is a design constraint rather than a result: any use that raises influence raises the credibility evidence required with it, to a level nothing in this paper approaches.

The later steps — the credibility assessment plan, its execution, the report, and the determination of adequacy for the context of use — all require qualification against *real* batch data from the specific process, product and site, which does not exist for this system. Alongside them, and under a risk-based framework such as GAMP 5 [15] and ICH Q9(R1) [14], a compliant system would need data integrity controls meeting ALCOA+ over its inputs, change control and revalidation triggers, drift monitoring with defined action limits, and a documented human review step the model cannot bypass. This paper contributes to none of them. It contributes a benchmark on which methods can be compared before any of that begins, and evidence that the comparisons currently being made are not informative.

8 Related work

Batch process monitoring. Multiway PCA and PLS for batch processes [21, 22] remain the reference method and the baseline we use. Simulation benchmarks in this area — PenSim [3] and the industrial-scale IndPenSim [10, 11] — are the established way to make the field reproducible, and our simulator follows their kinetic structure. Our disagreement is not with these benchmarks, which are careful about their own status, but with downstream work that treats performance on them as evidence about real processes.

Digital twins in pharmaceutical manufacturing. Reviews of the area [4] document a rapidly growing literature and, in our reading, a consistent gap: validation is nearly always against the model’s own simulation or against a single proprietary dataset that cannot be shared. The regulatory context for advanced manufacturing continues to develop [13], which makes the evidentiary standard more, not less, pressing.

Synthetic tabular data and its evaluation. CTGAN and TVAE [28] defined the modern baselines; sample-level fidelity metrics [1] and the utility-transfer framing used in healthcare [23]

shape how such models are assessed. Work on privacy is the closest in spirit to ours: Stadler et al. [25] show that synthetic data does not deliver the privacy guarantees commonly claimed for it, by testing the claim rather than assuming it. We do the analogous thing for the fidelity claim in a simulated-benchmark setting. Broader treatments of what synthetic data is for [16] make the same point we arrive at empirically: the purpose determines the evaluation, and an evaluation detached from the purpose is decorative. Generative models for medical time series [6] face a structurally identical problem.

9 Limitations

We list these in descending order of how much they should reduce confidence in the paper’s claims.

The real corpus is not pharmaceutical. SECOM is semiconductor manufacturing. It has the structure our protocol needs — real, high-dimensional process data with a rare, genuine disposition label — but the transfer result in §6.2 is, strictly, about transfer from our fermentation simulator to a semiconductor process. A defender of simulator-based validation can reasonably answer that a pharmaceutically faithful simulator would transfer better to a pharmaceutical corpus. We cannot rule that out, and no public data lets us test it. What the result does establish is that agreement between a simulator and real process data is not automatic and must be demonstrated rather than assumed. Obtaining even one real batch-record corpus, under any access arrangement, would be worth more than any methodological improvement in this paper. We tried for a closer substitute and did not get one: the IndPenSim 100-batch dataset [9] would have supplied pharma-relevant batch and phase structure, and it is not part of this study. It is also itself a simulation, so it would have narrowed the gap rather than closed it.

The simulator is not calibrated. Its structure follows published fed-batch models but its parameters were not fitted to plant data and its absolute product scale is arbitrary. Specification limits are consequently derived from its own nominal distribution (§4.2). Every monitoring and readiness number in §6 inherits this and describes the simulator only.

6 methods is a weak ranking test. Every correlation is a Spearman coefficient over six points. Computing it per seed gives 10 estimates and an interval, which is what we report, but it does not create information about methods we did not run: a seventh generator could sit anywhere. A larger and more diverse generator set would sharpen this. The per-seed estimates within a metric also share those six methods, so the Mann–Whitney contrast in §6.2 is descriptive rather than confirmatory.

The difficulty gap may be a property of our simulator, not of simulators. Our headline observation that the simulated task is far easier than the real one (0.893 ± 0.037 against 0.157 ± 0.022) could be read as a criticism of this particular simulator: perhaps a harder one would close the gap. We think that reading strengthens rather than weakens the paper’s point, since nobody building a simulator without real batch records can know where on that difficulty scale they have landed — but it does mean the specific magnitude of the gap is not a general constant, and we do not present it as one.

The VAE’s utility figure is not comparable. It rests on 4 of 10 seeds because the model failed to generate enough failing records in the rest (§6.3). We report it in Table 1 for completeness and exclude it from any claim about which method is best on real data.

The generators are re-implementations. We implement a WGAN-GP rather than running CTGAN itself, and a plain MLP VAE rather than TVAE. The families are represented; the specific published architectures and their tuned hyperparameters are not, and a generative-modeling reviewer would be right to want the reference implementations. Neural generators were trained with a fixed budget and no per-corpus hyperparameter search, which plausibly disadvantages them relative to the closed-form models.

The failure-mode prior is biased by construction. Recall records count escapes, not rejections. Modes that release reliably catches are under-represented in our prior, which means the simulator’s fault mix is skewed toward the failures that get through — arguably the wrong direction for a benchmark about detection.

We did not test whether the empirical prior changes anything. The abstract and §4.1 lead with the recall corpus as the study’s anchor to real data, and it does real work in one narrow sense: it fixes the relative frequency of six failure modes. It does not fix the absolute failure rate, the fault magnitudes, or the specification limits, all of which are assumed or self-derived. The obvious control — regenerate the corpus with a uniform failure-mode prior and see which conclusions move — was not run. Until it is, the influence of the anchoring on any reported result is unquantified, and the prominence we give it in the framing is not yet earned.

Privacy results are a demonstration, not a guarantee. Distance-based membership inference is a weak attack, the corpora contain no personal data, and the confidentiality interest in real batch records is commercial process knowledge rather than individual records. Nothing here licenses a privacy claim about a deployed system.

Monitoring uses one method family. Multiway PCA is the reference baseline, not the state of the art. A recurrent or attention-based detector might do better on the observable modes; it could not do better on the unobservable ones, which is where the ceiling lies.

10 Conclusion

Synthetic digital twins of pharmaceutical manufacturing are usually validated against the simulators that produced their training data. Running one identical evaluation protocol on a simulated corpus and on a real manufacturing corpus, we find that this practice is sound for some questions and empty for the one that matters. Fidelity and privacy rankings transfer well ($\rho \in [0.55, 0.97]$ over 6 metrics), which is more than we expected. Downstream utility rankings do not ($\rho = 0.09 \pm 0.31$ and $\rho = 0.14 \pm 0.35$; both intervals span zero), and the prediction task on our simulated corpus is so much easier than the real one (0.893 ± 0.037 against 0.157 ± 0.022) that absolute performance measured on it cannot be carried across at all — a gap that belongs to this simulator’s near-deterministic disposition label rather than to simulators in general, and whose size and sign a builder without real batch records has no way to check. Alongside this, generators misestimate the failing-batch rate by one to two orders of magnitude without their aggregate scores reflecting it, and trajectory monitoring has a hard ceiling set by failure modes the instrumentation cannot see.

The practical reading is narrow and, we think, actionable: a simulator is an acceptable venue for asking whether a generative model has learned a distribution, and is not evidence of anything about whether the resulting synthetic data is useful. We offer the benchmark, the protocol and the negative results as a starting point rather than a solution, and we are explicit that none of it belongs near a release decision. Batch release is a human act with a named owner and a legal consequence; the contribution available to modeling work here is to make the evidence for such systems checkable, which begins with declining to score them against themselves.

Reproducibility

All results are produced by the scripts in `src/`, run in order: `01_profile_real_data.py` profiles the real anchors, `02_simulator.py` generates the simulated corpus, `03_evaluate.py` runs the generator protocol on both corpora, `04_monitoring.py` runs the trajectory experiments, `05_figures.py` draws every figure, `06_tables.py` emits `numbers.tex` and the three `table_*.tex` files, `07_agreement_ci.py` computes the per-seed rank correlations, their intervals and the fidelity-versus-utility contrast, and `08_verify_manuscript.py` re-checks the manuscript against the recorded results. Every numeral in this paper is a macro defined in `numbers.tex` and read from a JSON file in `results/`; none is typed by hand. Seeds, library versions and the platform are recorded in each results file. All stochastic results use the same ten seeds. Claude (Anthropic) was used to assist in developing and revising the code in `src/`; every script was read, executed and validated by the author, and the recorded results files are the output of those validated runs.

Data availability

SECOM is available from the UCI Machine Learning Repository under a Creative Commons Attribution 4.0 International license, which permits reuse with attribution [20]; we use it unmodified and redistribute no part of it. The openFDA drug enforcement export is in the public domain [27]; the export used here is dated 2026-08-27, recorded in `results/failure_mode_prior.json`. The simulated corpus is not distributed because it does not need to be: script 02 regenerates it bit-for-bit from the master seed recorded in `results/simulator_spec.json`. No proprietary, confidential or internal data of any organization was used.

Code availability

All code, the recorded results files and the figure sources are available at <https://github.com/phanibalagam/circular-by-construction> and archived at <https://doi.org/10.5281/zenodo.22855260>, which resolves to the current version. Running the eight numbered scripts in `src/` in order reproduces every number and every figure in this paper from the two public datasets named above.

CRedit authorship contribution statement

Phani Kumar Balagam: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review and editing, Visualization, Project administration. Sole author; no funding was received and no other person contributed to this work.

A Simulator calibration record

§6 discloses that the simulator was calibrated before any evaluation was run. This appendix records the sequence, because a paper about undisclosed evaluation choices should let a reader audit its own. Nothing here is a result; all of it is modeling history.

What was set, and from where. Seventeen process parameters carry a nominal value and a batch-to-batch coefficient of variation, each tagged in `src/02_simulator.py` with the quantity it represents — Monod half-saturation constants for substrate and dissolved oxygen, biomass

and product yields, specific production rate, substrate inhibition of production, biomass decay, maintenance, product hydrolysis, oxygen transfer, feed and initial substrate concentrations, working volume and inoculum density. The coefficients of variation range from 0.02 (working volume) to 0.20 (inoculum density), the latter standing for seed-train variability. Nominal values sit in the ranges reported for the fed-batch penicillin model family this structure is taken from. None was fitted to data, from any plant, at any point.

The passes that were run. Kinetic parameters and the exponential feed profile were adjusted over several passes until two properties held: batches completed the production phase without exhausting substrate or oxygen, and the nominal titer distribution was unimodal with a spread that left a non-trivial fraction of batches near any plausible lower limit. Neither criterion refers to real data; both are internal consistency checks on the model. The number of passes was not logged, which is itself a limitation of this record.

The specification limits were changed once, and it mattered. Limits were first asserted as fixed values chosen to look like product specifications. Against the simulator’s own nominal output those values rejected every batch, which established only that the model’s absolute scales — titer above all — are arbitrary. They were replaced with percentiles of the simulator’s own nominal, fault-free distribution: the 0.27th percentile in each tail, the $\pm 3\sigma$ equivalent for a normal, estimated over 500 fault-free batches drawn on a random stream separate from the one that generates the study corpus. The discarded asserted values are not recorded and we do not reconstruct them. The consequence is stated where it matters rather than only here: a label defined by percentiles of the simulator’s own output means “predicting release readiness” on this corpus is predicting a label the simulator invented (§7), and every monitoring and readiness number inherits it (§9).

The failure rate is an assumption, not an estimate. The nominal batch failure rate was set to match SECOM’s observed rate so the two corpora are comparable in class balance. It was not taken from the recall corpus: recall records count market escapes, not batch rejections, and cannot estimate a rejection rate. The recall corpus fixes only the *relative* frequency of the six retained failure modes, after five openFDA categories were excluded with the reason for each recorded in the source. Fault magnitudes are hand-chosen uniform ranges. §9 states that the control for this — regenerating the corpus under a uniform failure-mode prior — was not run.

What was fixed in advance and not revisited. The master seed, the ten evaluation seeds, the generator set, the metric set, the four process phases, the eight release CQAs, the six-mode fault model and the two-of-six observability ceiling were all set before the evaluation and not changed after results arrived. The one budget change after that point is disclosed in §6: the neural and classifier-two-sample-test budgets were cut once for runtime, before the reported runs, and applied uniformly thereafter.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to support the drafting and editing of manuscript text and the development and revision of analysis code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. The author retained sole responsibility for all final decisions concerning the research question, study design, data selection, outcome definitions, analytical methods and experimental controls; independently verified all sources,

data and results; executed and validated the analyses; and interpreted the findings. No AI tool is listed as an author of this work, and none could be: authorship carries accountability that no tool can hold.

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